Corporate Finance

Lecture 9: Debt and Taxes

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Outline

A. The interest tax deduction (Chap. 15.1, 15.2)

B. Modigliani and Miller 2: debt, tax, and firm value (Chap. 15.2)

C. Modigliani and Miller 2: debt, tax and the cost of capital (Chap. 15.2)

D. Personal taxes (Chap 15.4)

E. Optimal capital structure with taxes (Chap. 15.5; elements of Chap.

16)

A. The interest tax deduction

- In most countries: interests = expenses
  - Interest expenses are thus (fully or partially) deductible
  - Remark: the dividends are paid from after-tax profits!
- Example:
  - France in 2026: marginal corporate tax rate = 25% $\Rightarrow$ interest expenses are deductible up to the higher of EUR 3 million or 30% of EBITDA.
  - US in 2026: marginal corporate tax rate (federal rate) = 21% $\Rightarrow$ interest expenses are deductible up to 30% of EBIT.

$\Rightarrow$ We will assume for simplicity that interest expenses are fully deductible.

Interest tax deduction: example

A company has no leverage and pays 34% tax on its profits of 10. The CEO decides to take on a loan of 80 with an interest of 5% and to reduce its equity by the same amount:

Without leverage:

Assets	Liabilities	
100	100	Equity
	0	Debt

With leverage:

Assets	Liabilities	
100	20	Equity
	80	Debt

	Without leverage	With leverage
Income before tax	10	10 - 4 = 6
Taxes	3.4	2.04
Net income	6.6	3.96
Interest paid to debt	0	4

Interest tax deduction: a 1st summary

- Leverage permits to reduce the tax payable:

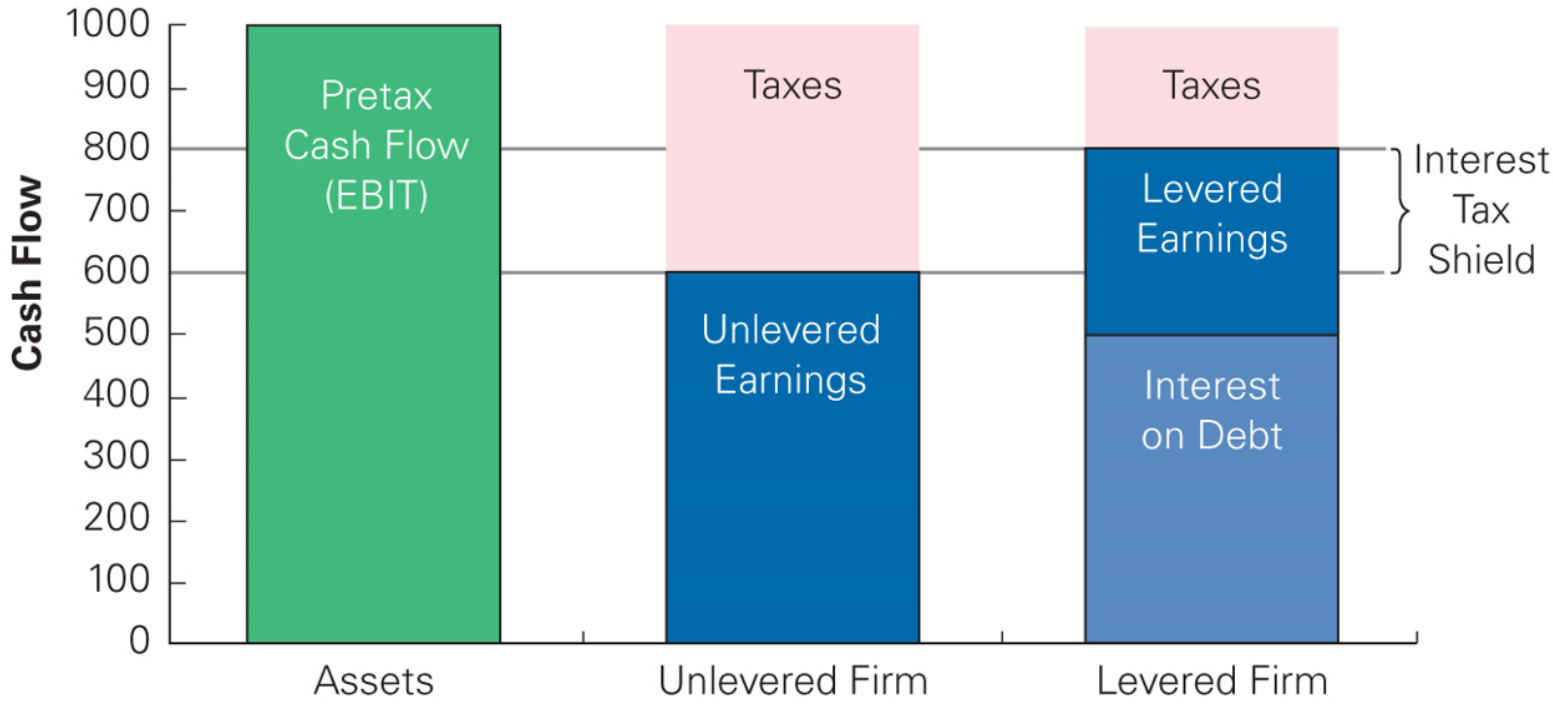
$$\text{Interest tax shield} = \text{Corporate tax rate} \times \text{Interest payments}$$

- Consequences $\Rightarrow$ increases revenues available for stock- and debtholders

$$\begin{pmatrix}\text{Cash Flows to Investors}\\ \text{with Leverage}\end{pmatrix} = \begin{pmatrix}\text{Cash Flows to Investors}\\ \text{without Leverage}\end{pmatrix} + \begin{pmatrix}\text{Interest Tax}\\ \text{Shield}\end{pmatrix}$$

- Thus: the total value of the company ($V = E + D$) increases.

Cash Flows of the unlevered and levered firm



B. MM2: debt, tax, and firm value

- Reminder: MM1 framework $\Rightarrow$ Perfect capital markets + no taxes
- MM2 $\Rightarrow$ introduction of corporate taxes

The total value of the levered firm exceeds the value of the firm without leverage due to the present value of the tax savings from debt:

$$V^L = V^U + PV(\text{Interest Tax Shield}).$$

MM2 and the value of the interest tax shield

- Hypothesis: the company has a perpetual and constant debt at the constant risk-free rate of r. Then:

$$\text{Annual tax shield} = \tau_C r D.$$

- Present value of the tax shield is:

$$PV(\text{Tax shield}) = \sum_{i=1}^{\infty} \frac{\tau_C r D}{(1+r)^i} = \frac{\tau_C r D}{1+r} + \frac{\tau_C r D}{(1+r)^2} + \frac{\tau_C r D}{(1+r)^3} + \ldots$$

  - The same saving $\tau_C r D$ every year, discounted one year further each time. Summing this perpetuity gives

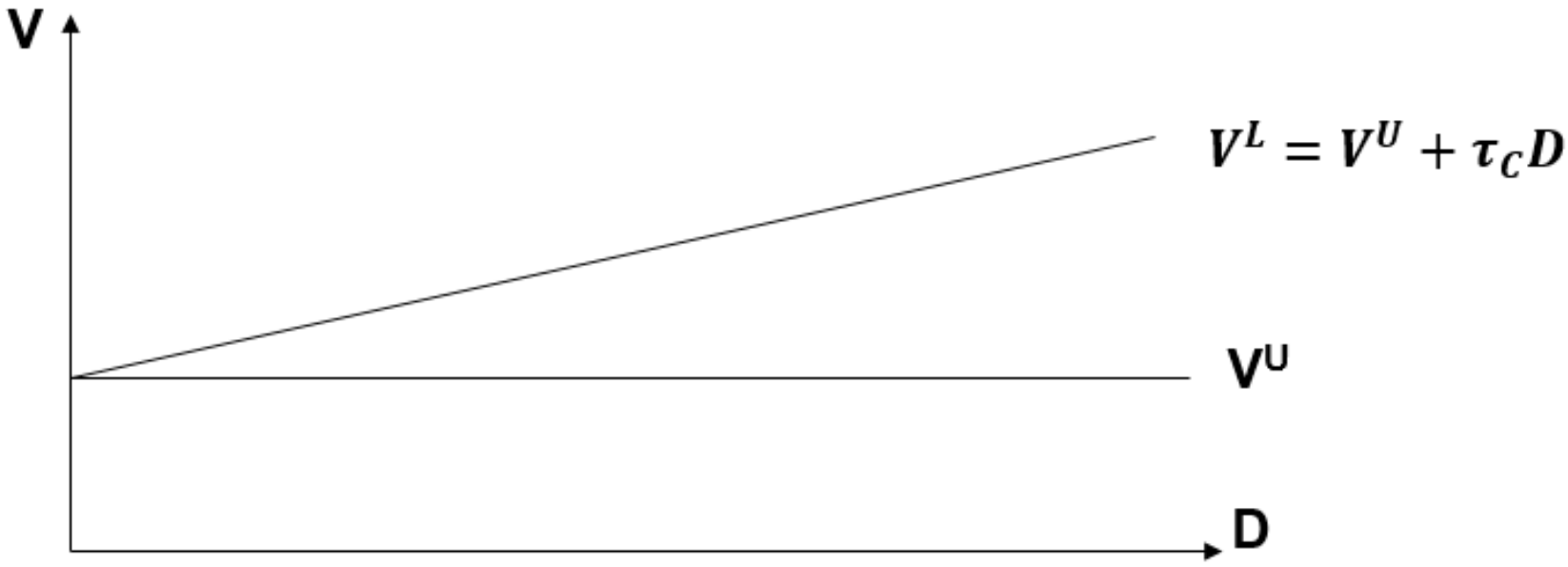
$$PV(\text{Tax shield}) = \frac{\tau_C r D}{r} = \tau_C D.$$

  - Remark: valid as long as the perpetuity has the same risk as the debt.

- The Proposition 1 of MM2 becomes then: $V^L = V^U + \tau_C D$.

MM2: a simple example

- Firm U is financed with 500 EUR of equity and has no leverage. The marginal corporate tax rate is 30%.
- What is the value of firm, L, which has 250 EUR debt and is otherwise identical?



C. MM2: debt, tax, and the cost of capital

Let us establish the relation between the cost of equity of the levered firm and the cost of equity of the unlevered firm:

$$r_E^L = \frac{(\Pi - r_D D)(1-\tau_C)}{E^L} = \frac{\Pi(1-\tau_C) - r_D D(1-\tau_C)}{E^L},$$

where Π is the operating income (EBIT). We know that

$$\Pi(1-\tau_C) = r_{WACC}^U V^U = r_E^U V^U,$$

and that

$$V^U = V^L - \tau_C D = (E^L + D) - \tau_C D.$$

Thus:

$$\begin{aligned} r_E^L &= \frac{r_E^U\left((E^L + D) - \tau_C D\right) - r_D D(1-\tau_C)}{E^L} \\ &= r_E^U + \frac{D}{E^L}(1-\tau_C)\left(r_E^U - r_D\right). \end{aligned}$$

Tax and the effective cost of debt

- Because debt creates tax savings, its effective cost is lower than the interest payments

$$\text{Effective cost of debt} = r_D\,(1-\tau_C)\,D.$$

- Example:
  - Debt of 100 000 EUR at the interest rate of 10%, with corporate tax rate of 34%.
  - Tax savings = 0.34*(0.1*100000) = 3400
  - Effective cost of debt = 0.1*100000 - 3400 = 6600

MM2: WACC with taxes

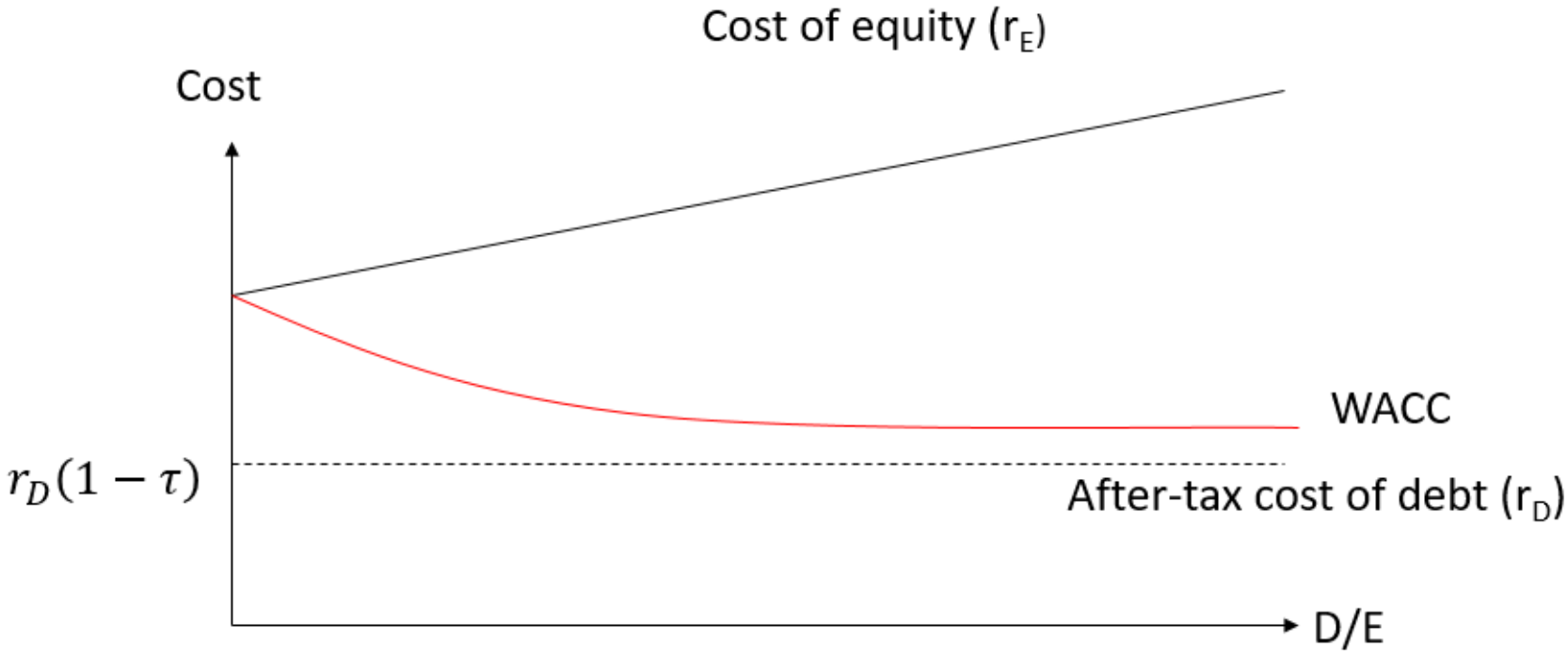
- The WACC of a levered firm is:

$$r^L_{WACC} = \frac{D}{D+E^L} r_D (1-\tau_C) + \frac{E^L}{D+E^L} r^L_E,$$
$$= \frac{D}{D+E^L} r_D + \frac{E^L}{D+E^L} r^L_E - \frac{D}{D+E^L} \tau_C r_D.$$

- By using the expression in Equation [eq:1] for r^L_E, we obtain:

$$r^L_{WACC} = \underbrace{r^U_{WACC}}_{\substack{\text{WACC without} \\ \text{leverage}}} \underbrace{\left(1 - \tau_C \frac{D}{D+E^L}\right)}_{\substack{\text{Reduction due} \\ \text{to Interest Tax Shield}}}$$

MM2: Summary in a graph



- Remark: In MM2, we have (this can be derived using the expression in Equation [eq:1]):

$$\beta_E^L = \beta_E^U \left(1 + (1-\tau_C)\,\frac{D}{E^L}\right).$$

Beyond MM2: Tax shield and "uncertain'' debt levels

- MM2 $\Rightarrow$ perpetual, constant debt
  - Discounting the tax shield at the risk-free rate
- If the future level of leverage is unknown, a higher discount rate has to be used.
  - Remark: "general" case $\Rightarrow$ $V^L = V^U + PV(\text{Interest Tax Shield})$.
- "Special" case: target debt-equity ratio
  - We can then evaluate the company of the levered company, V^L, by discounting the FCF "to the Firm" at the after-tax WACC.
  - The value of the interest tax shield is then the difference between V^L and the unlevered value of the company, V^U, that can be found by discounting the FCF at the pre-tax WACC.

Recapitalizing to capture the tax shield: example

- Micron SA has no debt and 20 million shares outstanding with a market price of 15 EUR per share. The company pays a 33% marginal tax rate.
- The CEO has decided to borrow 100 million EUR on a permanent basis through a leveraged recap in which borrowed funds would be used to repurchase outstanding shares.
  - What is the impact on firm value?

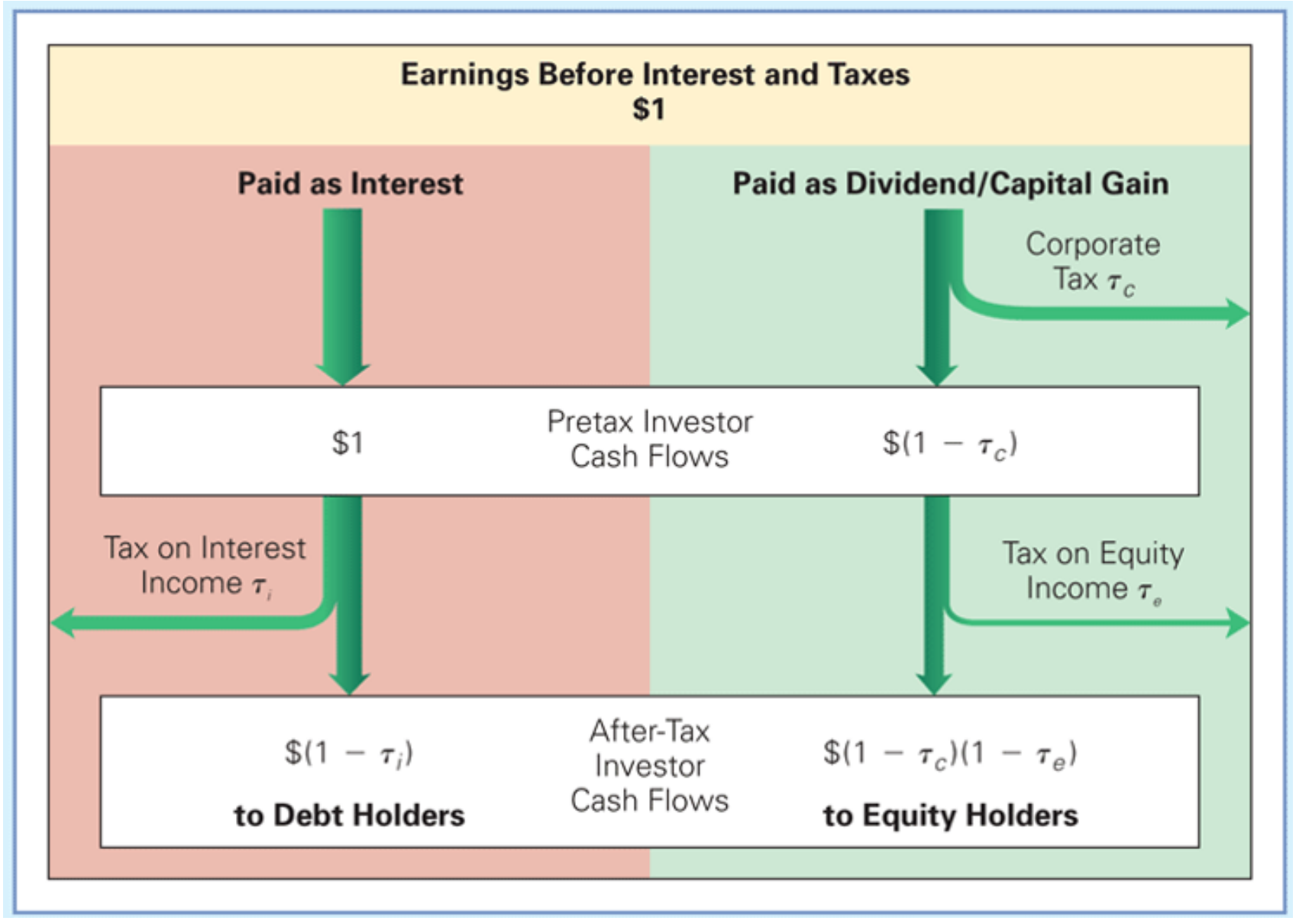
Recapitalizing to capture the tax shield: example (con't)

		Step 1	Step 2	St
	Initial	Recap Announced	Debt Issuance	Sh
Assets (in M EUR)				
V(unlevered firm)	300	300	300	30
Interest tax shield	0	33	33	33
Cash	0	0	100	0
Total assets	300	333	433	33
Liabilities (in M EUR)				
Equity	300	333	333	23
Debt	0	0	100	10
Total liabilities	300	333	433	33
Shares outstanding (M)	20	20	20	13
Price per share (EUR)	15	16.65	16.65	16

D. Personal taxes

- See: Merton MILLER "Debt and taxes", Journal of Finance, 1977
- Simple observation: between 1930 and 1950, the marginal company tax rate has increased from 10% to 52% in the US ... but the liabilities of US companies have not changed significantly.
- Explanation: MM2 overstates the tax savings due to leverage
  - Personal tax rates need to be taken into account!
  - Reminder: investors also need to pay income taxes (dividends, capital gains, and interest income).

After-Tax Investor Cash Flows Resulting from $1 in EBIT

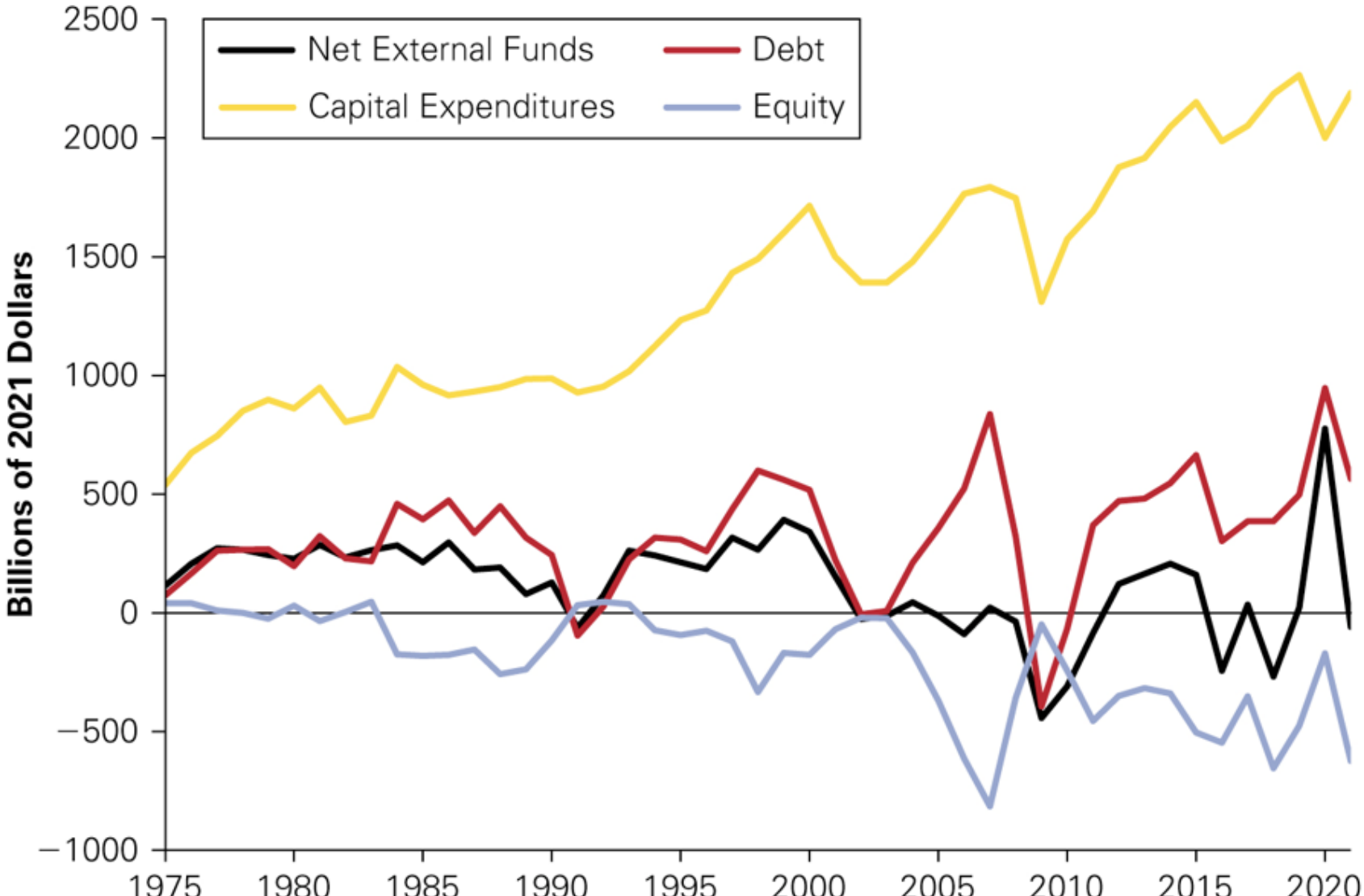


E. Optimal capital structure with taxes

- Question(s):
  - What leverage maximizes firm value?
  - What leverage minimizes the WACC?
- Reminder: firm value is the sum of all FCF "to the Firm" (which do not depend on leverage) discounted at the WACC.
- Thus, the leverage which minimizes the WACC also maximizes firm value.

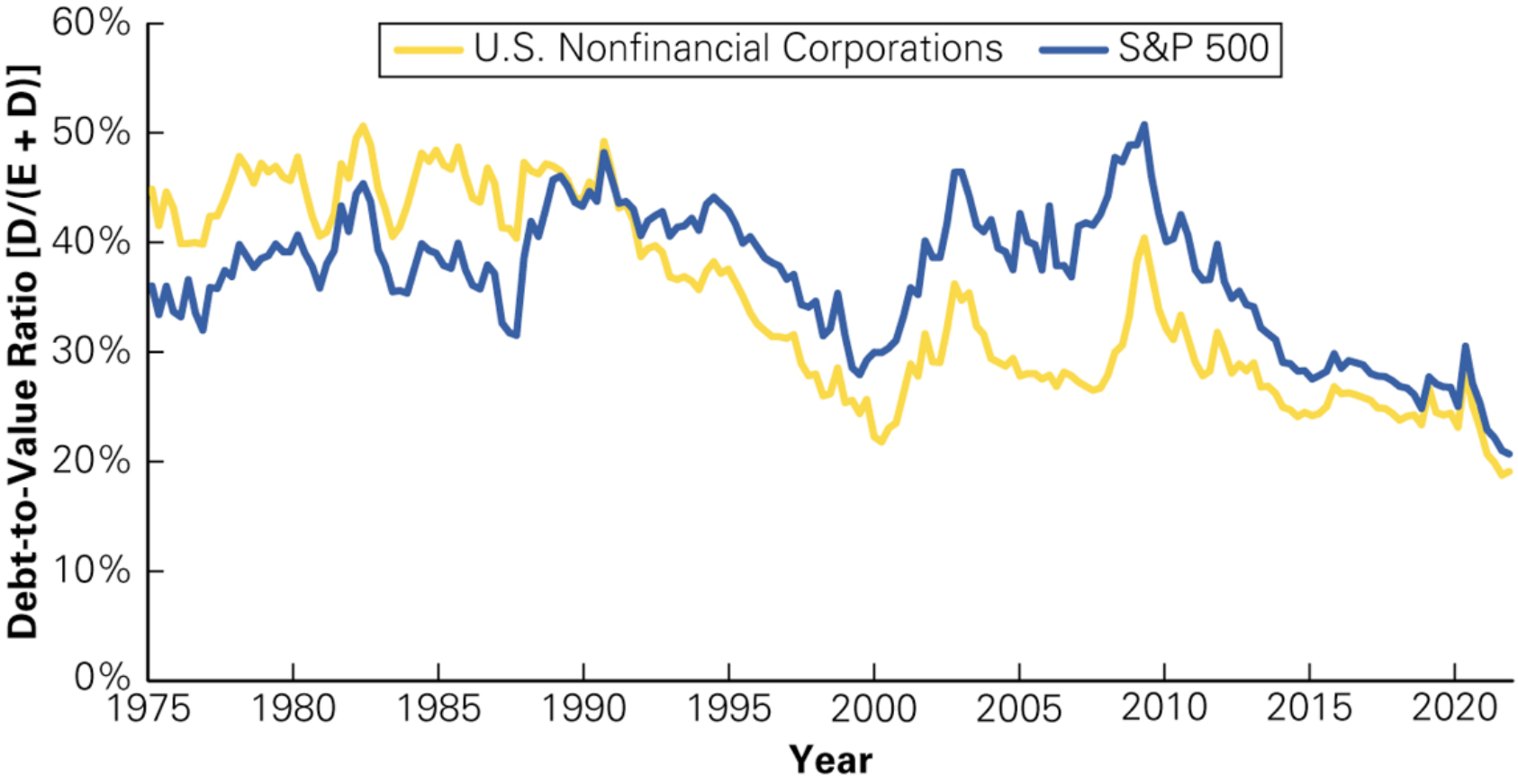
Do firms prefer debt?

Net External Financing and Capital Expenditures by U.S. Corporations, 1980-2021



Do firms prefer debt? (con't 2)

Debt-to-Value Ratio $[D/(E+D)]$ of U.S. Firms, 1975-2021



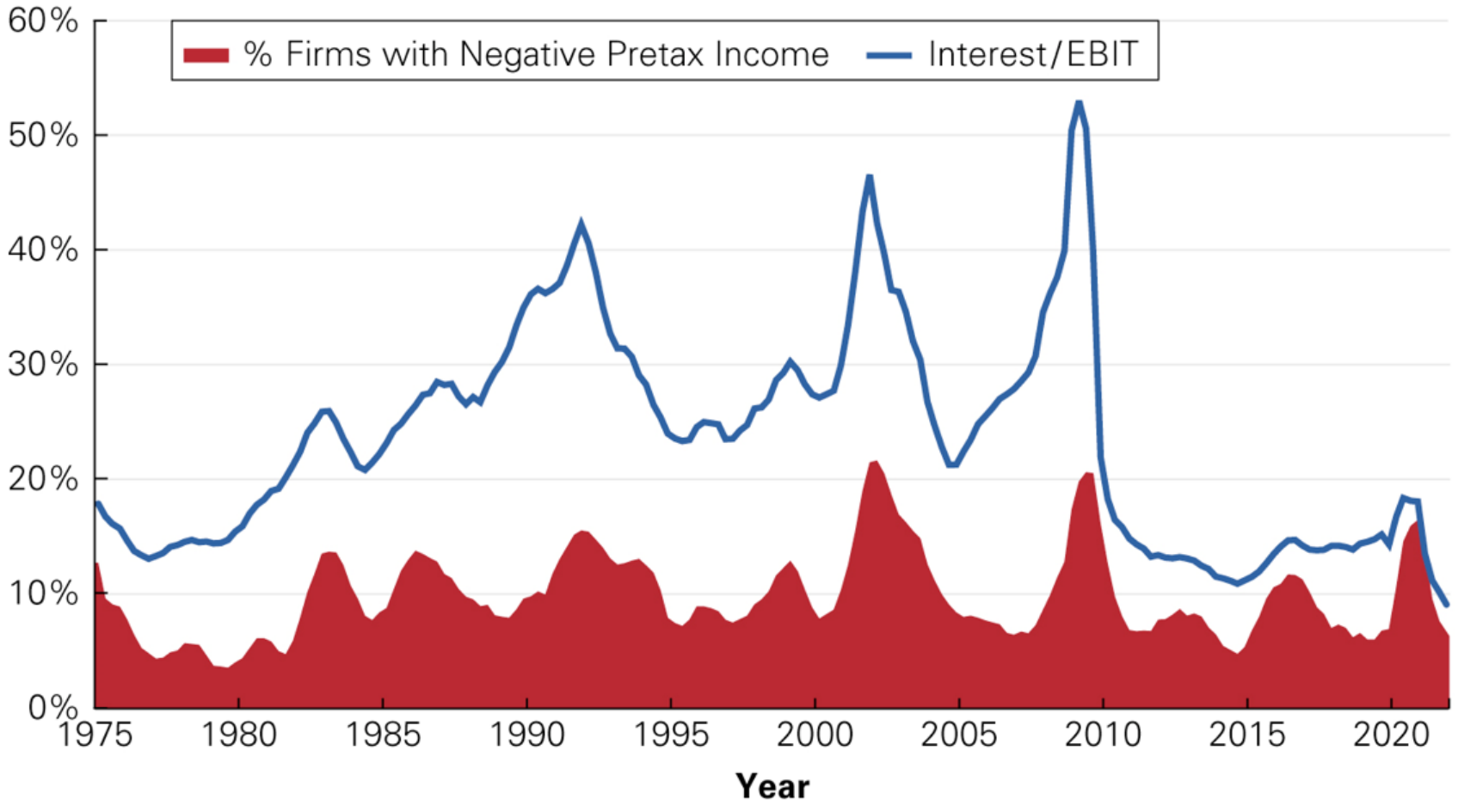
- Source: Compustat and Federal Reserve, Flow of Funds Accounts of the United States, 2021.
- Remark: large companies only

Limits to the tax benefit of debt

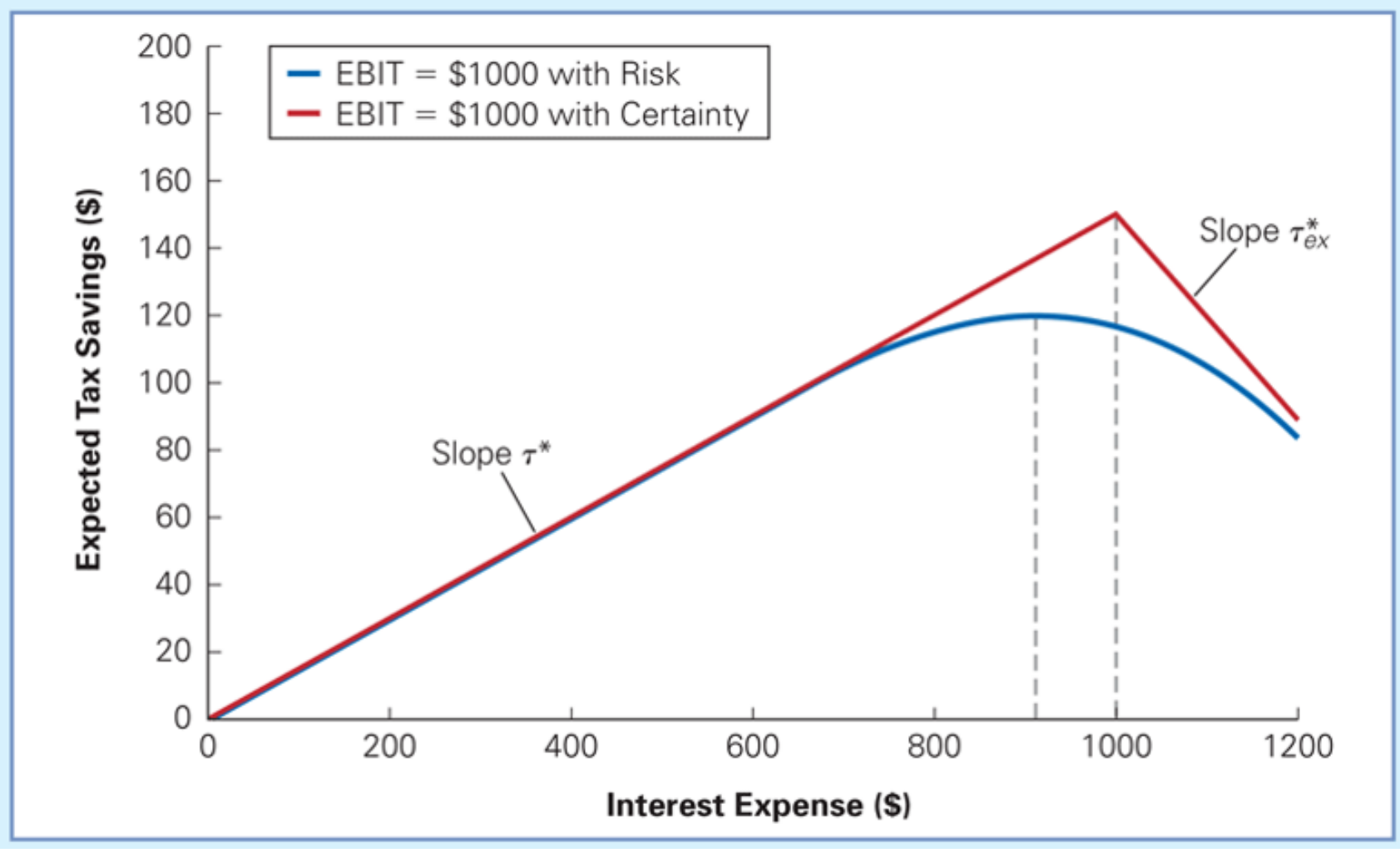
- In 2020, debt represented on average "only" 20% of the liabilities of large U.S. companies.
  - Remark: large variations by industry.
  - Average leverage of for French firms similar.
- Why does debt make up less than half of the capital structure of most firms?
- Why does the leverage choice vary so much across industries?

The low leverage puzzle

Interest Payments as a Percentage of EBIT and Percentage of Firms with Negative Pre-tax Income, S&P 500 1975-2021



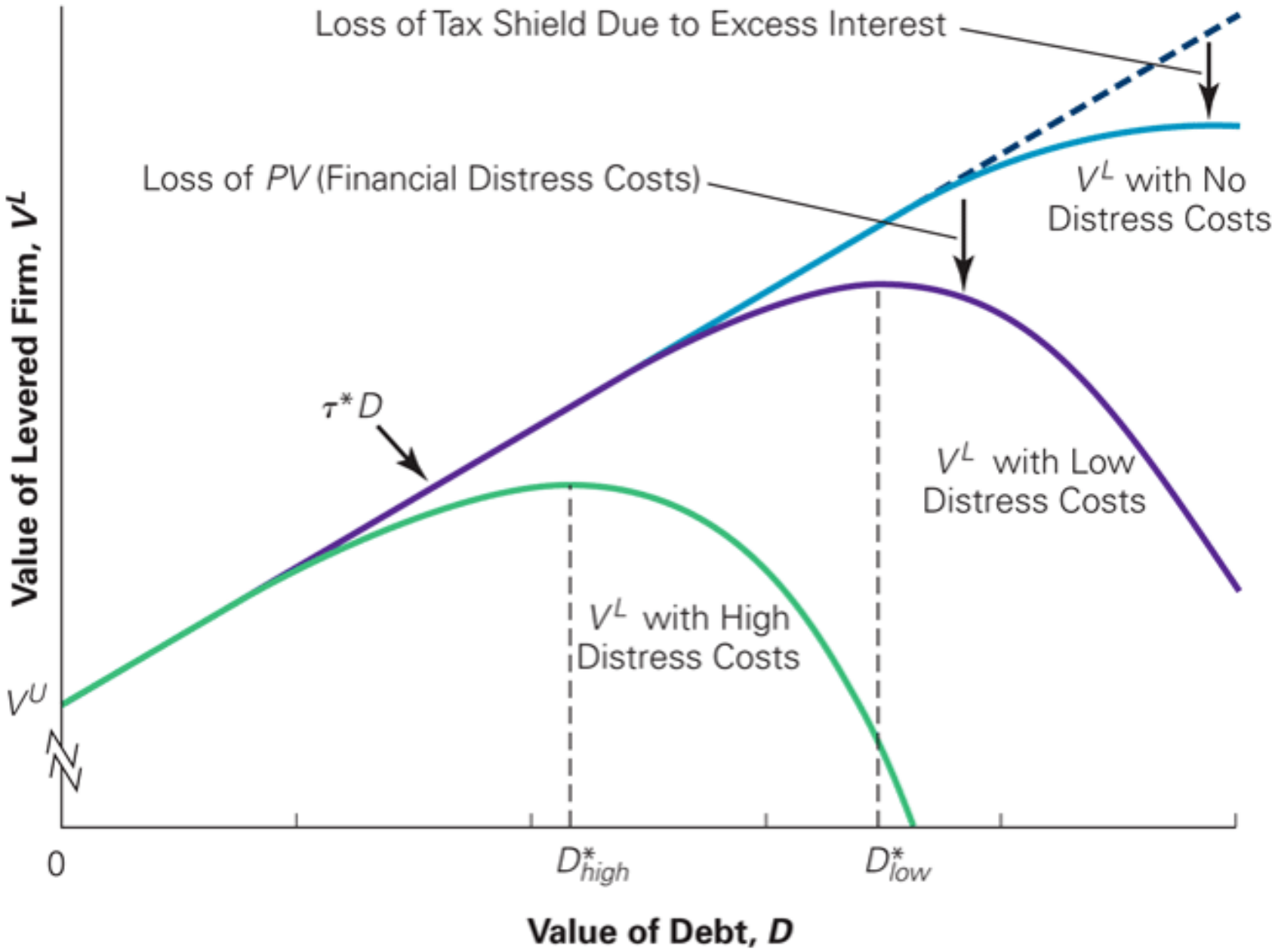
Tax savings for different levels of interest



How to explain the "low'' leverage?

- A number of factors not taken into account in Modigliani and Miller explain the low observed levels of leverage:
  - Bankruptcy costs: the possibility of bankruptcy doesn't change MM's arguments but if there are costs associated to bankruptcy (direct or indirect), the value of cash flows available for investors decreases.
  - Agency costs: the capital structure changes the way the company is managed and thus the FCF (contradicting MM's hypotheses). This effect can be positive or negative.

Optimal leverage with taxes and financial distress costs



Other theories of optimal capital structure

- Trade-off theory
  - Optimal capital structure weighs the benefits of debt that result from shielding cash flows from taxes against the costs of financial distress associated with leverage.
- "Pecking order theory" (linked to signalling)
  - Managers prefer to use retained earnings first, then debt and will issue equity only as a last resort.